REPUBLIC POLYTECHNIC

C339 Software Testing and Analytics
Continuous Assessment 1 (CA1)

Project 01: Shop Order Calculation System Requirements Specification

1. About

A retail shop requires a simple software system to calculate the total cost of purchased items and determine whether an order qualifies for free delivery.

The system consists of two independent functions:

- **Calculate Total:** Calculates the total purchase amount based on the unit price and quantity purchased.
- **Free Delivery Eligibility:** Determines whether an order qualifies for free delivery based on its total purchase amount.

Both functions must operate according to the requirements described below.

2. General Requirements

The following requirements apply to both functions:

1. All monetary amounts are represented in **Singapore cents (SGD)**, using whole numbers without decimal places.
2. All numeric inputs must be Python integers (int). Boolean values (True and False) are not considered valid integers for this system.
3. If an argument has an invalid data type, the function must raise a **TypeError**.
4. If an argument has the correct data type but falls outside the permitted range, the function must raise a **ValueError**.
5. The functions operate independently and do not require a database, network connection, or shared application state.

3. Functional Requirements

3.1 Calculate Total Purchase Amount

Source File: cart_total.py

Function: calculate_total(unit_price_cents, quantity)

Description

The function calculates the total purchase amount for a single product based on its unit price and the quantity purchased.

Parameter	Data Type	Constraints
unit_price_cents	int	Must be a non-negative integer
quantity	int	Must be a non-negative integer

Requirements:

1. The total purchase amount must be calculated by multiplying the unit price by the quantity purchased.
2. A unit price or quantity of zero results in a total purchase amount of zero.
3. **Output:** The function must return the calculated total purchase amount expressed in cents.

3.2 Determine Free Delivery Eligibility

Source File: cart_delivery.py

Function: qualifies_free_delivery(total_cents)

Description:

The function determines whether an order qualifies for free delivery based on its total purchase amount.

Parameter	Data Type	Constraints
total_cents	int	Must be a non-negative integer

Requirements:

1. Orders with a total purchase amount of S$50.00 or more qualify for free delivery.
2. Orders below S$50.00 do not qualify for free delivery.
3. Delivery eligibility depends solely on the total purchase amount. Customer membership status does not affect eligibility.

4. **Output:** The function returns a Boolean. True means the order qualifies for free delivery while False means the order does not qualify for free delivery.